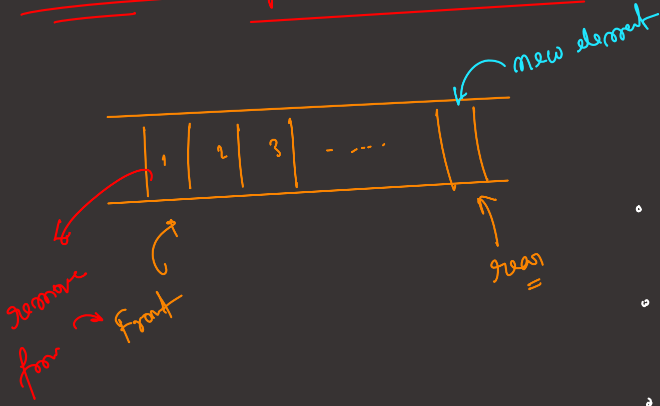
Queue   First in First Out

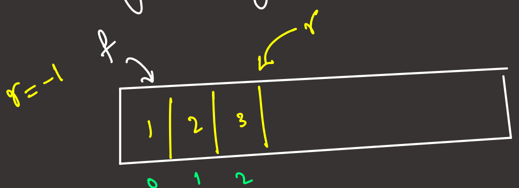
insertion order = deletion order



- insert / add / enqueue $O(1)$
- deletion / remove / dequeue . $O(1)$ Deque ← DS
- Peak () → Front $O(1)$

Implementation of Queue

(i) using Array

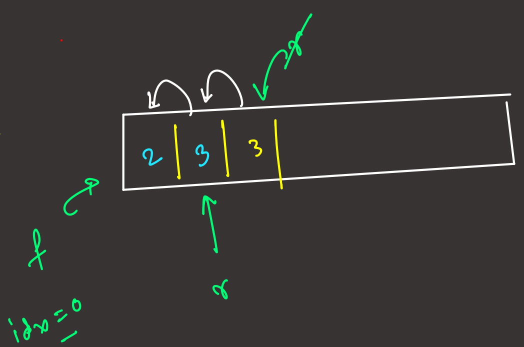


$O(1)$

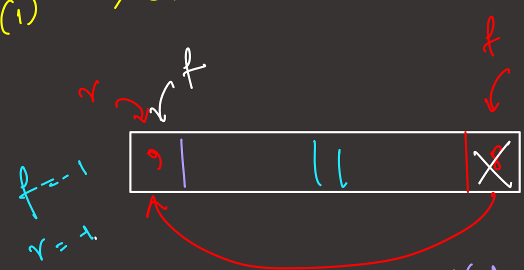
insert

[1, 2, 3]

remove



$O(n)$ → $O(1)$ → Circular Queue



add(9)

add (1)
add (2)
add (3)
remove () → 1

rear = (rear + 1) % size
front = (front + 1) % size

full → { r - f = size - 1 }

$(r+1) \% \text{size} = f$

condition of empty

{ f = -1
r = -1 }